

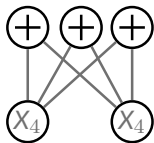
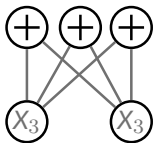
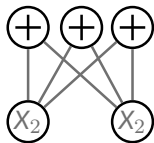
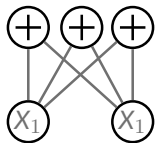
Positive Unital Circuits

Pedro Zuidberg Dos Martires

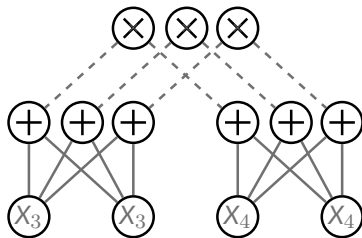
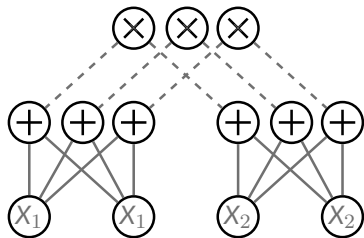
Probabilistic Circuits



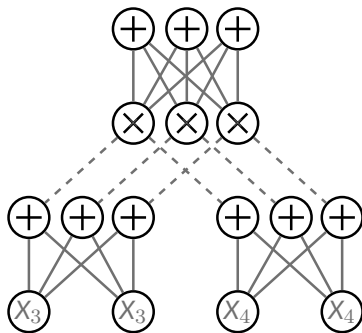
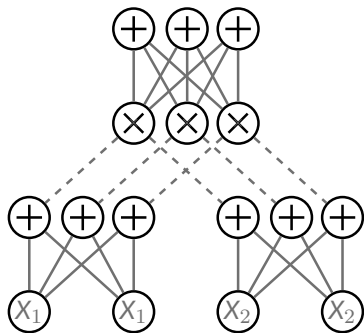
Probabilistic Circuits



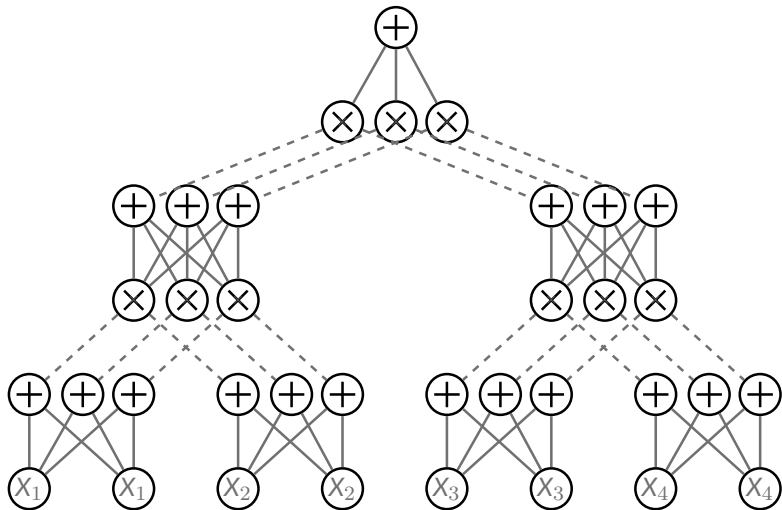
Probabilistic Circuits



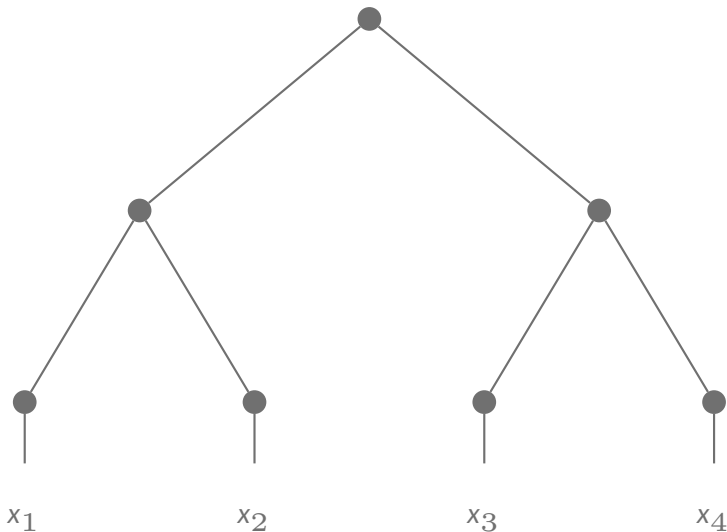
Probabilistic Circuits



Probabilistic Circuits



Abstracted Probabilistic Circuit a.k.a Tensor Trees



Computation Units

$$P_k(\mathbf{x}_k) = \begin{cases} P_{x_k} & \text{if } k \text{ is leaf} \\ W_k \times (P_{k_l}(\mathbf{x}_{k_l}) \otimes P_{k_r}(\mathbf{x}_{k_r})) & \text{else} \end{cases}$$

Computation Units

$$P_k(\mathbf{x}_k) = \begin{cases} P_{x_k} & \text{instantiate vector} \\ W_k \times (P_{k_l}(\mathbf{x}_{k_l}) \otimes P_{k_r}(\mathbf{x}_{k_r})) & \text{matrix-vector product} \end{cases}$$

When do we have a probability distribution?

$$\sum_{x_k} P_{x_k} = [1, \dots, 1]^T$$

$$\forall k, i, j : W_{kij} \geq 0$$

$$\forall k, i : \sum_j W_{kij} = 1$$

Monotone and Non-Monotone Circuits

Monotone circuits are not fully expressive¹

Non-monotone circuits (negative weights):
NP-hard to check if positive function²³

¹Honghua Zhang, Steven Holtzen, and Guy Van den Broeck (2020). “On the relationship between probabilistic circuits and determinantal point processes”. In: *UAI*. PMLR, pp. 1188–1197.

²Juha Harviainen, Vaidyanathan Peruvemba Ramaswamy, and Mikko Koivisto (2023). “On inference and learning with probabilistic generating circuits”. In: *UAI*. PMLR, pp. 829–838.

³Sanyam Agarwal and Markus Bläser (2024). “Probabilistic Generating Circuits-Demystified”. In: *Forty-first International Conference on Machine Learning*.

But quantum mechanics does computations with complex numbers and always outputs a probability!

Can we use this to construct non-monotone circuits?

What Do We Need for a Probability Distribution?

$$\Omega(X) = \{x_1, \dots, x_N\}$$

Associate to each outcome x_i a probability $p(x_i)$.

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$$\forall x : p(x) \geq 0 \quad (\text{positivity})$$

$$\sum_x p(x) = 1 \quad (\text{completeness})$$

We need one positive number for each event.

Positive Operator Valued Measures (POVM)

$$\Omega(X) = \{x_1, \dots, x_N\}$$

Associate to each outcome x_i an operator $O(x_i) \in \mathbb{C}^{N \times N}$.

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$O(x_i)$ is positive-semi definite

$(v^* O(x_i) v \geq 0, \text{ for every } v \in \mathbb{C}^N)$

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Completeness:

$$\sum_{x_i \in \Omega(X)} O(x_i) = \mathbb{1}$$

We need one PSD matrix for each event.

Boundedness

Probabilities:

$$\forall x : p(x) \leq 1 \quad (1)$$

POVMs:

$$\forall x : O(x) \leq \mathbb{1} \quad (2)$$

Follows from positivity and completeness.

Obtaining Probabilities from a POVM⁴⁵

We need a “density matrix” ρ .

ρ is a positive semidefinite matrix.

ρ has unit trace ($\text{Tr}[\rho] = 1$).

⁴Lev Landau (1927). “Das dämpfungsproblem in der wellenmechanik”. In: *Zeitschrift für Physik* 45.5, pp. 430–441.

⁵John von Neumann (1927). “Wahrscheinlichkeitstheoretischer Aufbau der Quantenmechanik”. In: *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse* 1927, pp. 245–272.

Obtaining Probabilities from a POVM⁴⁵

We need a “density matrix” ρ .

ρ is a positive semidefinite matrix.

ρ has unit trace ($\text{Tr}[\rho] = 1$).

$$p(x) = \text{Tr}(E_x \rho)$$

Each event is characterized by a positive semidefinite matrix E_x .

This is guaranteed to produce a valid probability distribution.

⁴Lev Landau (1927). “Das dämpfungsproblem in der wellenmechanik”. In: *Zeitschrift für Physik* 45.5, pp. 430–441.

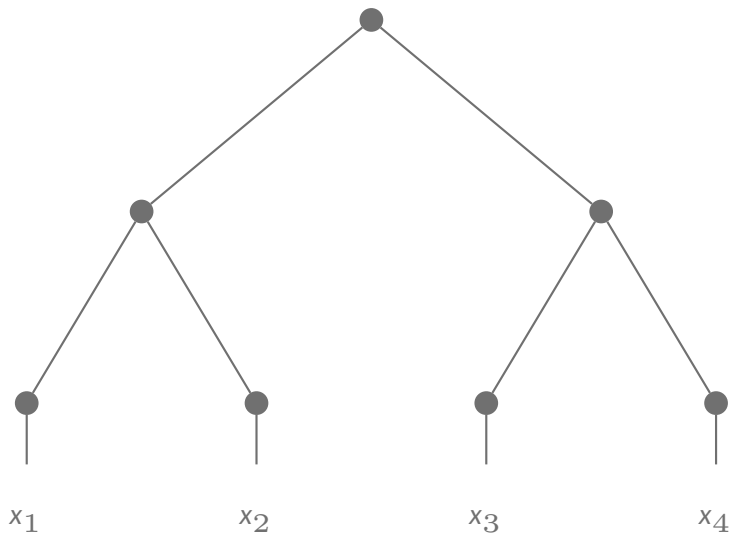
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Let's represent probabilistic events using a circuit.

How can we construct events from “sub-events”?

We need to make sure that computation units preserve PSD.

Abstracted Probabilistic Circuit a.k.a Tensor Tree



Quantum Computation Units

$$O_k(\mathbf{x}_k) = \begin{cases} E_{x_k} & \text{if } k \text{ is leaf} \\ \Phi_k(O_{k_l}(\mathbf{x}_{k_l}) \otimes O_{k_r}(\mathbf{x}_{k_r})) & \text{else,} \end{cases}$$

$$\Phi(Q(\mathbf{x})) = \sum_{j=1}^D K_j Q(\mathbf{x}) K_j^*$$

Kraus' Theorem⁶

$$Q(\mathbf{x}) \in \mathbb{C}^{N \times N}$$

$$Q'(\mathbf{x}) \in \mathbb{C}^{M \times M}$$

$$Q'(\mathbf{x}) = \sum_{j=1}^{j \leq N \times M} K_j Q(\mathbf{x}) K_j^*$$

⁶K Kraus (1983). "States, Effects, and Operations Fundamental Notions of Quantum Theory". In: *Lecture Notes in Physics* 190

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$$Q'(\mathbf{x}) = \sum_{j=1}^{j \leq N \times M} K_j Q(\mathbf{x}) K_j^*$$

$$\sum_j K_j^* K_j \leq \mathbb{1}$$

“universality”: any quantum computation can be expressed using the of Kraus operators $\{K_j\}_{j=1}^{j \leq N \times M}$

“non-uniqueness”: the Kraus operators are not unique for a given quantum operation
(existence of unitary equivalence)

⁶K Kraus (1983). “States, Effects, and Operations Fundamental Notions of Quantum Theory”. In: *Lecture Notes in Physics* 190

Kronecker products conserve PSD.
(Hadamard products are a special case)

Quantum operations conserve PSD.

$$p(\mathbf{x}) = \text{Tr}[O_{\text{root}}(\mathbf{x})\rho]$$

$$p(\mathbf{x}) = \text{Tr}[\mathbf{O}_{root}(\mathbf{x})\rho]$$

We have a parametrized non-monotone function.

Guaranteed to have probability distribution over $\mathbf{x}$.

We retain tractable marginalization!

Generalize computations closed over
positive reals
to computations closed over
positive semidefinite matrices.

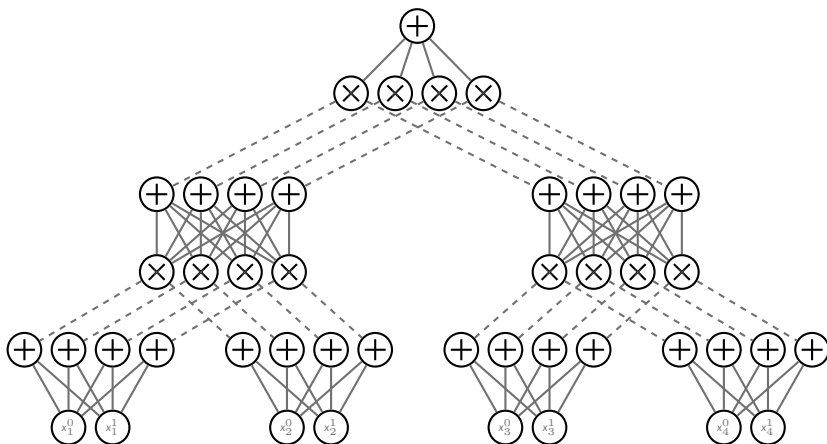
Pedro Zuidberg Dos Martires (2025). "A Quantum Information Theoretic Approach to Tractable Probabilistic Models". In: *Proceedings of the Uncertainty in Artificial Intelligence Conference*

Establishes connection between tractable models and quantum information theory.

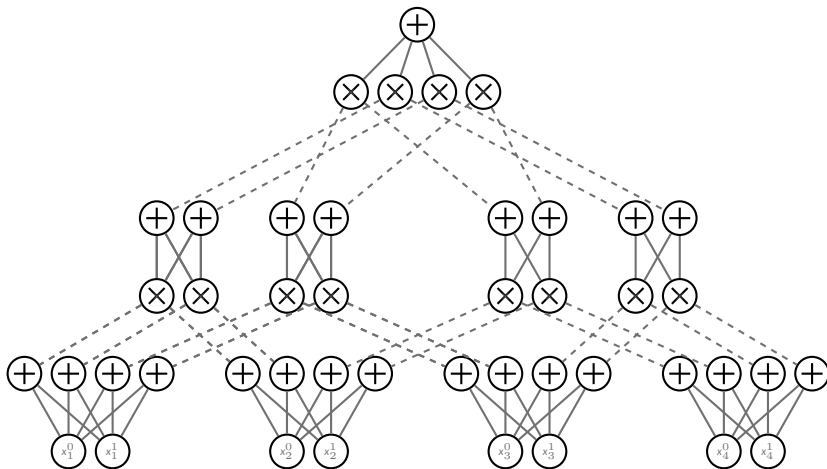
Establishes a model class only using quantum operations (no hacks).

Probabilistic circuits, squared-circuits, and Born machines follow as special cases.

Going Away from Trees (structured decomposability) ...



...To DAGs (non-structured decomposability)



Tensorized Computation in Tree-Like Structures

$$O_k(\mathbf{x}_k) = \begin{cases} E_{x_k} & \text{if } k \text{ is leaf} \\ \Phi_k(O_{k_l}(\mathbf{x}_{k_l}) \otimes O_{k_r}(\mathbf{x}_{k_r})) & \text{else,} \end{cases}$$

Node-Wise Computation in Tree-Like Structures

$$O_k(\mathbf{x}_k) = \begin{cases} E_{x_k} & \text{if } k \text{ leaf unit} \\ O_{k_l}(\mathbf{x}_{k_l}) \otimes O_{k_r}(\mathbf{x}_{k_r}) & \text{if } k \text{ product unit} \\ \Phi_k(O_k(\mathbf{x}_j)) & \text{if } k \text{ sum unit} \end{cases}$$

Node-Wise Computations in DAG-Like Structures

$$\mathbf{O}_k(\mathbf{x}_k) = \begin{cases} E_{\mathbf{x}_k} & \text{if } k \text{ leaf unit} \\ \mathbf{O}_{k_l}(\mathbf{x}_{k_l}) \otimes \mathbf{O}_{k_r}(\mathbf{x}_{k_r}) & \text{if } k \text{ product unit} \\ \sum_{j \in \text{in}(k)} w_{kj} \Phi_{kj}(\mathbf{O}_j(\mathbf{x}_j)) & \text{if } k \text{ sum unit} \end{cases}$$

with $w_{kj} \geq 0$ and $\forall k : \sum_j w_{kj} = 1$

We now sum over different ways of partitioning the variables in the sum unit. (convex combination of PSD matrices is PSD)

Conclusion

PUnCs generalize many (all?) multi-linear tractable models.

Computation gates based on quantum operations from quantum information theory

Unclear connection to orthogonal circuits (non-structured decomposable) by Loconte, Javaloy, and Vergari⁷

⁷Lorenzo Loconte, Adrián Javaloy, and Antonio Vergari (2025). “How to Square Tensor Networks and Circuits Without Squaring Them”. In: *arXiv preprint arXiv:2512.17090*.